CHAPTER 04

# Activities & Lab Packet

## Ecology & Ecosystems

A complete set of labs, worksheets, writing prompts, discussions, and a group project to accompany Chapter 4. Works with both the Standard and Easy Read versions of the textbook.

📗 **Georgia Standard S7L4**

### WHAT'S INSIDE

# 1 Build-a-Food-Web Lab

LAB

**OBJECTIVE**
Build a food web with yarn and explain how energy flows and how organisms depend on one another.

**TIME**
40 min

**STANDARDS**
S7L4

## Materials (per group)

- Organism cards (sun, grass, rabbit, grasshopper, mouse, snake, hawk, fox, mushroom/decomposer)
- 1 ball of yarn
- Tape and a large open space or table

## Setup

Spread the cards out. Each student or pair holds one organism card. The sun starts with the yarn.

## Student Instructions

1. The sun passes the yarn to every producer (who makes its own food).
2. Each organism passes yarn to whatever eats it — energy flows from food to feeder. Keep holding your strand.
3. When the web is built, the teacher names an organism that "disappears." That person tugs their strands — everyone who feels a tug is affected.
4. Record which organisms were affected and explain the chain reaction.

⚠️ **TEACHER NOTES**

**The big idea:** removing ONE organism tugs the whole web — everything is connected. Decomposers should connect back to producers (recycling nutrients). Have the yarn always travel *from the eaten to the eater* so arrows = direction of energy.

## Food-Web Lab — Data Sheet

Name: ______________________ Period: ____________

| Organism | Producer / Consumer / Decomposer | What it eats | What eats it |
|---|---|---|---|
| Grass | | | |
| Rabbit | | | |
| Hawk | | | |
| Mushroom | | | |

If all the hawks disappeared, predict TWO things that would happen to the rest of the web:

## 2 Ecosystem in a Jar

LAB

**OBJECTIVE**
Build a small closed ecosystem and observe how biotic and abiotic parts interact over time.

**TIME**
30 min setup + 2 weeks observation

**STANDARDS**
S7L4

### Materials (per group)

- Clear jar with lid
- Gravel, soil, small plants or seeds (producers)
- Water, and optionally small organisms (e.g., a snail or pillbugs)

### Student Instructions

1. Layer gravel, then soil. Plant seeds or small plants.
2. Add water and (optionally) a small consumer. Seal the lid and place in indirect light.
3. List the biotic (living) and abiotic (non-living) parts you included.
4. Observe for 2 weeks. Record changes in water (condensation), plant growth, and any organisms.

⚠️ **TEACHER NOTES**

**Connect it:** the sealed jar recycles water (evaporation → condensation) and gases (photosynthesis ↔ respiration). No new matter enters, but energy (light) does. Great lead-in to the water and carbon cycles.

### Ecosystem Jar — Observation Log

Name: ______________________ Period: ____________

| Biotic parts | Abiotic parts |
|---|---|
| | |

| Day | What I observed |
|---|---|
| 1 | |
| 4 | |
| 7 | |
| 14 | |

Where did the water go and come back from inside the sealed jar?

## 3 Energy Pyramid & the 10% Rule

LAB

**OBJECTIVE**
Model the loss of energy at each level of a food chain and explain why top predators are rare.

**TIME**
30 min

**STANDARDS**
S7L4

### Materials (per group)

- 1000 dried beans (or paper squares) = energy units from the sun's producers
- 4 cups labeled: Producers, Primary, Secondary, Tertiary

### Student Instructions

1. Put all 1000 beans in the Producers cup — that's the energy plants captured.
2. Move 10% (100 beans) to Primary consumers. The other 90% was "used up" as heat and life activities.
3. Move 10% again (10 beans) to Secondary consumers, then 10% again (1 bean) to Tertiary.
4. Build a pyramid drawing showing how much energy is left at each level.

### Energy Pyramid — Data Sheet

Name: ____________________ Period: ____________

| Level | Example organism | Energy units left | % of original |
|---|---|---|---|
| Producers | grass | 1000 | 100% |
| Primary consumer | | | 10% |
| Secondary consumer | | | 1% |
| Tertiary consumer | | | 0.1% |

Use your data to explain why there are far fewer hawks than there are grasshoppers in a field:

**ANSWER KEY**
Only ~10% of energy passes to the next level; the rest is lost as heat and used for life processes. So each level up has much less energy available, supporting fewer organisms. Top predators need a huge base of producers beneath them.

## 4 Biotic vs. Abiotic

WORKSHEET

**OBJECTIVE**
Sort the living and non-living parts of an ecosystem and explain how they interact.

**TIME**
15 min

**STANDARDS**
S7L4

Name: ______________________ Period: _____________

Part A: Write B for biotic (living/once-living) or A for abiotic (non-living).

| Item | B / A | Item | B / A |
|---|---|---|---|
| Oak tree | | Sunlight | |
| Rock | | Earthworm | |
| Water | | Mushroom | |
| Temperature | | Fallen leaf | |
| Soil | | Beetle | |

Part B: Short answer.

1. Give one way an abiotic factor (like rainfall) can affect a biotic factor. ______________________________
2. Why is a dead log considered biotic even though it isn't alive anymore? ___________________________
3. List the levels of organization from smallest to largest: organism, ___, ___, ___. ______________________________

**ANSWER KEY**

Part A: Oak B, Sunlight A, Rock A, Earthworm B, Water A, Mushroom B, Temperature A, Fallen leaf B, Soil A (mostly), Beetle B. Part B: 1) less rain → fewer plants → less food for animals; 2) it was once living and still came from an organism; 3) organism › population › community › ecosystem.

## 5 Food Webs & Energy Flow

WORKSHEET

| OBJECTIVE | TIME | STANDARDS |
|---|---|---|
| Read a food web to identify roles and trace the flow of energy. | 20 min | S7L4 |

Name: ______________________   Period: _____________

Use this food chain: **Sun** → **Grass** → **Grasshopper** → **Frog** → **Snake** → **Hawk**

Part A: Label each organism's role.

| Organism | Producer / Consumer type |
|---|---|
| Grass | |
| Grasshopper | |
| Frog | |
| Hawk | |

Part B: Answer.

1. Which way do the arrows point — toward the food or toward the eater? _________
2. Where does ALL the energy in this chain originally come from? _________
3. What role recycles dead organisms back into the soil? _________
4. If grass died off, name TWO organisms directly hurt. ________________________

Part C: Draw an arrow food chain using: owl, mouse, seeds, sun.

**ANSWER KEY**

Part A: Grass producer; Grasshopper primary consumer (herbivore); Frog secondary consumer; Hawk tertiary consumer / top predator. Part B: 1) toward the eater (direction of energy); 2) the sun; 3) decomposers; 4) grasshopper and (indirectly) frog/snake/hawk. Part C: Sun → seeds → mouse → owl.

# 6 Biomes of the World

WORKSHEET

| OBJECTIVE | TIME | STANDARDS |
| --- | --- | --- |
| Match major biomes to their climate, plants, and animals. | 20 min | S7L4 |

Name: ____________________ Period: ____________

Part A: Match the biome to its description.

| Biome | Description |
| --- | --- |
| Desert | |
| Tropical rainforest | |
| Tundra | |
| Grassland | |
| Ocean | |

*Clues:* very dry, little rain • warm & wet, most species • frozen, treeless, short growing season • open land, grazing herds • largest biome, covers most of Earth

Part B: Short answer.

1. What two abiotic factors are most used to classify biomes? ______________________
2. Why do few large trees grow in the tundra? ______________________
3. Name a physical adaptation that helps an animal survive in the desert. ______________________

**ANSWER KEY**

Part A: Desert = very dry; Rainforest = warm & wet, most species; Tundra = frozen, treeless; Grassland = open land, grazing herds; Ocean = largest biome. Part B: 1) temperature and precipitation/rainfall; 2) frozen ground (permafrost) and short growing season; 3) e.g., storing water, nocturnal habits, thick skin.

## 7 Writing Prompts

WRITING

| OBJECTIVE | TIME | STANDARDS |
| --- | --- | --- |
| Use scientific vocabulary to explain ecological relationships in writing. | 15–30 min each | Cross-cutting |

### Level 1 — Beginner (1 paragraph)

1. **A Day in the Web.** Pick one animal. Describe what it eats and what eats it, using the words producer, consumer, and energy.
2. **Living or Not.** Describe your favorite outdoor place and list three biotic and three abiotic parts of it.

### Level 2 — Intermediate (3 paragraphs)

3. **The Missing Link.** Choose an organism and explain what would happen to its ecosystem if it disappeared. Trace the chain reaction through at least three other organisms.
4. **Follow the Energy.** Explain the 10% rule and use it to argue why a food chain rarely has more than four or five links.

### Level 3 — Advanced (essay)

5. **Invader!** Research or imagine an invasive species entering a new ecosystem. Construct an argument explaining how a single new organism can disrupt food webs, limiting factors, and native populations — and propose one solution.

## 8 Classroom Discussion Questions

DISCUSSION

**OBJECTIVE**
Practice scientific reasoning through think-pair-share.

**TIME**
Warm-ups, exit tickets, seminar

**STANDARDS**
Cross-cutting

**⚠️ TEACHER NOTES**
Think-pair-share format. Several questions have no single right answer — push for evidence from the chapter.

### Quick Warm-Ups (1–2 min)

1. Is the sun biotic or abiotic? Why is it still essential to every food web?
2. Could an ecosystem survive with no decomposers? What would pile up?
3. Why are producers always at the bottom of the energy pyramid?
4. Name a limiting factor that could stop a population from growing forever.
5. Are you a producer, consumer, or decomposer? Explain.

### Deeper Discussions (5–15 min)

6. Why does removing a top predator sometimes cause MORE problems than removing a plant?
7. If energy is lost at every level, why doesn't an ecosystem just "run out" of energy? Where is it constantly resupplied?
8. Matter (like carbon and water) is recycled, but energy is not. Explain the difference.
9. Two ecosystems get the same sunlight but one is a desert and one is a rainforest. What abiotic factor explains the difference?
10. Humans are part of ecosystems too. Give one example of a limiting factor that affects human populations.

## 9 Design an Ecosystem Challenge

GROUP

| OBJECTIVE | TIME | STANDARDS |
| --- | --- | --- |
| Design a balanced ecosystem and defend why it would stay stable. | 2 class periods | S7L4 |

### Project Description

Each group invents an ecosystem — real or imaginary — and proves it is balanced by showing how energy and matter flow through it. Present as a poster, model, or slideshow.

### Project Requirements

#### The Setting

- A biome with its climate (temperature & rainfall)
- At least 4 abiotic factors
- A name and a map or drawing

#### The Living World

- ≥ 2 producers, 3 consumers, 1 decomposer
- A food web diagram with energy arrows
- One limiting factor that keeps it in balance

### The Twist (presentation day)

The teacher introduces a change — a drought, a new predator, or an invasive species. Each group must predict and explain how their ecosystem responds.

⚠️ TEACHER NOTES

**Grading:** accurate roles & energy flow 35%, complete food web 25%, response to the twist 20%, creativity 10%, presentation 10%.
**Extension:** have groups trade "twists" with each other.

## 10 Vocabulary Bingo

REVIEW

| OBJECTIVE | TIME | STANDARDS |
| --- | --- | --- |
| Review chapter vocabulary in a low-stress game. | 20–30 min | S7L4 |

### Setup

Blank 5×5 grid; word list on board; FREE middle. Read a clue, students mark the word. 5 in a row wins.

### Vocabulary List (24 words)

| | | | | |
| --- | --- | --- | --- | --- |
| ecosystem | biotic | abiotic | producer | consumer |
| decomposer | herbivore | carnivore | omnivore | food chain |
| food web | energy pyramid | photosynthesis | predator | prey |
| population | community | habitat | limiting factor | carrying capacity |
| biome | niche | nutrient cycle | 10% rule | — |

### Sample Clue Cards (read aloud, don't show)

| Clue to read aloud | Answer |
| --- | --- |
| An organism that makes its own food using sunlight | producer |
| A non-living part of an ecosystem, like sunlight or water | abiotic |
| An organism that breaks down dead matter and returns nutrients | decomposer |
| All the feeding relationships in an area, connected together | food web |
| An animal that only eats plants | herbivore |
| Anything that limits how big a population can grow | limiting factor |
| The largest number of organisms an area can support | carrying capacity |
| A large region defined by its climate, plants, and animals | biome |
| The animal that is hunted and eaten | prey |
| Only about 10% of energy passes to the next level — this is the… | 10% rule |
| All the members of one species living in an area | population |
| An organism that eats both plants and animals | omnivore |

⚠️ TEACHER NOTES

**Easy adaptation:** give the list with definitions during play. Run multiple patterns so several students win.